

My Magic UW – The Art of the Break

Checked quotes, unchecked sentences

Badger BuildFest 2026 Live stress test on 27 September 2026 github.com/benverhaalen/magic-uw

1. THE AGENT

DEFINITION 1 (Grounded answer). My Magic UW is a desktop study app for UW–Madison students. A student asks a question about a course. The student’s own AI (Claude Code or Codex) writes the answer as short sentences $s_1, \dots, s_n$. Each sentence s_i cites a quote q_i from the course materials P . Code checks the answer before the student sees it. Our design rule is “AI writes, code decides.”

DEFINITION 2 (The old check). The function `checkAnswer` accepted a sentence when its quote was real, so $\text{accept}(s_i) \iff q_i \subseteq P$. It never compared the sentence s_i with its quote q_i .

2. THE FAILURE

CLAIM 1. The old check shows a false sentence to the student when the sentence cites a real quote. The real citation makes the false fact look more trustworthy.

Proof by an example from the live test. The student asks “How much is the final project worth?” Our agent cites the real quote $q =$ “final project 45%”, so $q \subseteq P$. Its sentence starts “The syllabus originally set the final project at 30%”, and it corrects itself later in the same sentence. The old check accepts this sentence, because it tests only $q \subseteq P$. The student sees 30% next to a working citation. ■

REMARK. We searched our code for other checks that prove the evidence but not the conclusion. We found three more. (i) An email with a course code in its subject got the label “Course staff” from any sender. (ii) A Jev judgment (a hosted AI model) could raise any email to urgent, with no limit. (iii) A link that named the course number became the official course site, also when a student posted it in a discussion.

3. THE STRESS TEST

DEFINITION 3 (Live test). We wrote 20 synthetic course scenarios to cause the failure. The five types are a changed syllabus date, room or weight, numbers near each other, arithmetic, weekdays, and two courses on one page. The student’s Claude Code (`claude-sonnet-5`) answered each scenario 3 times through the real prompt, schema and client path of the app. We ran this on two builds, the first fix and the final fix. Code marked each sentence, and we read every sentence ourselves. In each run, 6 of the 60 answers correctly said “Not in your materials” and had no sentences.

DEFINITION 4 (Targeted test). For each of the four weak checks, we generated 60 cases with a scripted fake model. Each case contains the trigger, so this test measures the fix and not the natural rate.

4. HOW OFTEN IT HAPPENED

PROPOSITION 1. Our agent wrote 1 sentence that contradicts its own cited quote in 167 sentences (0.6%). The old check shows it with a working citation. Before the fixes, each weak check failed in 60 of 60 targeted cases. Brackets give Wilson 95% intervals.

Live test on our agent	First fix (87 sentences)	Final fix (80 sentences)
Sentences that contradict their own cited quote	0/87	1/80 [0.2, 6.7]% and caught
Contradicting sentences that the student sees	0/87	0/77 [0, 4.8]%
Answers where the check replaced a correct sentence	15/60 [15.8, 37.2]%	2/60 [0.9, 11.4]%
Targeted test, 60 cases for each weak check	Before	After
Wrong sentence shown next to a real quote	60/60	0/60 [0, 6.0]%
Each of the three other weak checks (mail and site links)	60/60 each	0/60 each
Correct results kept (staff mail urgent, real sites found)	60/60	60/60

5. WHAT WE DID

DEFINITION 5 (First fix, `0f710ed`). Let $c(s)$ be the set of dates, weekdays, numbers and names in a text s . The new check `claimsMatch` requires

$$\text{accept}(s_i) \iff q_i \subseteq P \text{ and } c(s_i) \subseteq c(q_i).$$

If s_i fails, the app shows the quote q_i word for word with its link. The student loses the wording but not the fact.

OBSERVATION. The first fix passed every synthetic test. The live test then broke it. It replaced a correct sentence in 15 of 60 answers, and all 17 marked sentences were correct. The causes were numbers that the agent calculated (“175 points”), labels just outside the quote (“Homework 4”), and the room “B10” read as a name.

DEFINITION 6 (Final fix, `459669c`). A calculated number $z \in c(s_i)$ is accepted only when s_i shows the arithmetic $x \circ y = z$ with $x, y \in c(q_i)$, and code computes $x \circ y$ again. An example is “ $8 - 1 = 7$, so $7 \times 25 = 175$ points”. The prompt now tells the agent to show this. False rejects fell from 15/60 to 2/60.

REMARK. We also fixed the three other checks. A course code in a subject now gives only the label “Mentions *course*”. Only known staff addresses can make mail urgent. A Jev judgment raises mail by one level at most. A link only in a discussion never becomes the course site.

6. WHAT WE LEARNED

COROLLARY 1. A verifier must check the claim and not only the evidence. A real citation on a wrong sentence is worse than no citation.

COROLLARY 2. A check that passes on failures that we wrote can fail on real output. Ours passed every synthetic test and then rejected a correct sentence in 1 of 4 live answers. Measure the cost of a fix on the real agent.

COROLLARY 3. Calculated claims need their own path. The AI shows the arithmetic, and code does it again.

COROLLARY 4. When code confirms something, ask what it really proved. Each of our four checks proved less than the decision that used it.

Limits. All pages are synthetic, so we used no student data and no real course data. The live test has 60 answers from one model on one day.

Evidence. Links go to one fixed commit. See the live runner `run.ts`, the raw answers for the **final fix** and the **first fix**, the targeted cases `break-cases.ts`, and the full write-up `break-card.md`.

Rerun. `pnpm exec tsx evals/break-live/run.ts 3 5` and `pnpm exec tsx --test tests/break-*.test.ts`.